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Nonrepetitive colorings of graphs excluding a fixed immersion or topological minor

Paul Wollan¹ | David R. Wood² 

¹Department of Computer Science,
University of Rome “La Sapienza,”
Rome, Italy

²School of Mathematical Sciences,
Monash University, Melbourne,
Victoria, Australia

Correspondence

David R. Wood, School of Mathematical
Sciences, Monash University, Melbourne,
Victoria 3800, Australia.

Email: david.wood@monash.edu

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Abstract

We prove that graphs excluding a fixed immersion have bounded nonrepetitive chromatic number. More generally, we prove that if H is a fixed planar graph that has a planar embedding with all the vertices with degree at least 4 on a single face, then graphs excluding H as a topological minor have bounded nonrepetitive chromatic number. This is the largest class of graphs known to have bounded nonrepetitive chromatic number.

KEYWORDS

graph coloring, immersion, nonrepetitive coloring, topological minor

1 | INTRODUCTION

A vertex coloring of a graph is *nonrepetitive* if there is no path for which the first half of the path is assigned the same sequence of colors as the second half. More precisely, a k -coloring of a graph G is a function ψ that assigns one of k colors to each vertex of G . A path $(v_1, v_2, \dots, v_{2t})$ of even order in G is *repetitively* colored by ψ if $\psi(v_i) = \psi(v_{t+i})$ for $i \in \{1, \dots, t\}$. A coloring ψ of G is *nonrepetitive* if no path of G of even order is repetitively colored by ψ . Observe that a nonrepetitive coloring is *proper*, in the sense that adjacent vertices are colored differently. The *nonrepetitive chromatic number* $\pi(G)$ is the minimum integer k such that G admits a nonrepetitive k -coloring. We only consider simple graphs with no loops or parallel edges.

The seminal result in this area is by Thue [41], who in 1906 proved that every path is nonrepetitively 3-colorable. Thue expressed his result in terms of strings over an alphabet of three characters—Alon et al [3] introduced the generalization to graphs in 2002. Nonrepetitive graph colorings have since been widely studied [2–12, 21, 25–33, 35, 37–39]. The principle result of Alon et al [3] was that graphs with maximum degree Δ are nonrepetitively $\mathcal{O}(\Delta^2)$ -colorable. Several subsequent papers improved the constant [16, 26, 30]. The best-known bound is due to Dujmović et al [16].

Theorem 1 (Dujmović et al [16]). *Every graph with maximum degree Δ is nonrepetitively $(1 + o(1))\Delta^2$ -colorable.*

A number of other graph classes are known to have bounded nonrepetitive chromatic number. In particular, trees are nonrepetitively 4-colorable [8,33], outerplanar graphs are nonrepetitively 12-colorable [5,33], and graphs with bounded treewidth have bounded nonrepetitive chromatic number [5,33]. (See Section 2 for the definition of treewidth.) The best-known bound is due to Kündgen and Pelsmayer [33].

Theorem 2 (Kündgen and Pelsmayer [33]). *Every graph with treewidth k is nonrepetitively 4^k -colorable.*

The primary contribution of this paper is to provide a qualitative generalizations of Theorems 1 and 2 via the notion of graph immersions and excluded topological minors.

A graph G contains a graph H as an *immersion* if the vertices of H can be mapped to distinct vertices of G , and the edges of H can be mapped to pairwise edge-disjoint paths in G , such that each edge vw of H is mapped to a path in G whose endpoints are the images of v and w . The image in G of each vertex in H is called a *branch vertex*. Structural and coloring properties of graphs excluding a fixed immersion have been widely studied [1,13,14,18–20,22–24,34,36,40,42]. We prove that graphs excluding a fixed immersion have bounded nonrepetitive chromatic number.

Theorem 3. *For every graph H with t vertices, every graph that does not contain H as an immersion is nonrepetitively $4^{t^4 + O(t^2)}$ -colorable.*

Since a graph with maximum degree Δ contains no star with $\Delta + 1$ leaves as an immersion, Theorem 3 implies that graphs with bounded degree have bounded nonrepetitive chromatic number (as in Theorem 1).

We strengthen Theorem 3 as follows (although without explicit bounds). A graph G contains a graph H as a *strong immersion* if G contains H as an immersion, such that for each edge vw of H , no internal vertex of the path in G corresponding to vw is a branch vertex.

Theorem 4. *For every fixed graph H , there exists a constant k , such that every graph G that does not contain H as a strong immersion is nonrepetitively k -colorable.*

Note that planar graphs with n vertices are nonrepetitively $\mathcal{O}(\log n)$ -colorable [15], and the same is true for graphs excluding a fixed graph as a minor or topological minor [17]. It is unknown whether any of these classes have bounded nonrepetitive chromatic number. Our final result shows that excluding a special type of topological minor gives bounded nonrepetitive chromatic number. A *subdivision* of a graph H is a graph obtained from H by replacing each edge vw of H by a path with endpoints v and w . A graph G contains H as a *topological minor* if a subdivision of H is a subgraph of G . Vertices with degree at least 4 are important for topological minors since it is easily seen and well known that for a graph H with maximum degree 3, a graph G contains H as a topological minor if and only if G contains H as a minor.

Theorem 5. *Let H be a fixed planar graph that has a planar embedding with all the vertices of H with degree at least 4 on a single face. Then there exists a constant k , such that every graph G that does not contain H as a topological minor is nonrepetitively k -colorable.*

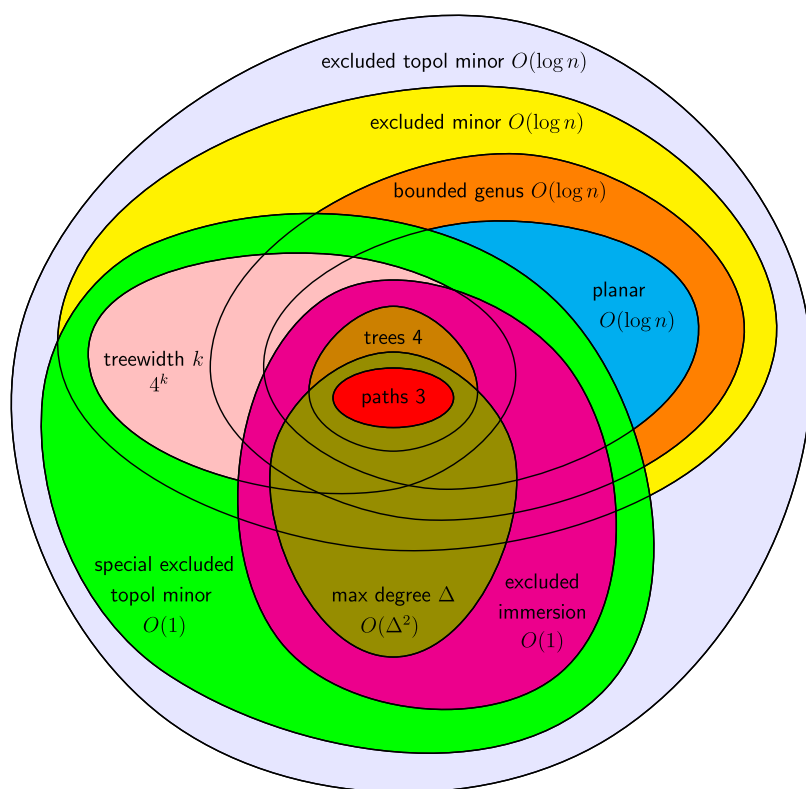


FIGURE 1 Upper bounds on the nonrepetitive chromatic number of various graph classes. “Special” refers to the condition in Theorem 5 [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jgt.2240)]

Graphs with bounded treewidth exclude fixed walls as topological minors. Since walls are planar graphs with maximum degree 3, Theorem 5 implies that the graphs of bounded treewidth have bounded nonrepetitive chromatic number (as in Theorem 2). Similarly, for every graph H with t vertices, the “fat star” graph (which is the 1-subdivision of the t -leaf star with edge multiplicity t) contains H as a strong immersion. Since fat stars embed in the plane with all vertices of degree at least 4 on a single face, Theorem 5 implies that graphs excluding a fixed graph as a strong immersion have bounded nonrepetitive chromatic number (as in Theorem 4). In this sense, Theorem 5 generalizes all of Theorems 1 to 4.

The results of this paper, in relation to the best-known bounds on the nonrepetitive chromatic number, are summarized in Figure 1.

Finally, note that several papers study nonrepetitive choosability. In particular, all of the $O(\Delta^2)$ upper bounds mentioned above hold for nonrepetitive choosability. Contrarily, Fiorenzi et al [21] showed that trees have unbounded nonrepetitive choosability. It follows that for all of the above graph classes with unbounded degree, the nonrepetitive choosability is unbounded.

2 | TREE DECOMPOSITIONS

For a graph G and tree T , a *tree decomposition* or *T -decomposition* of G consists of a collection $(T_x \subseteq V(G) : x \in V(T))$ of sets of vertices of G , called *bags*, indexed by the nodes of T , such that

for each vertex $v \in V(G)$ the set $\{x \in V(T) : v \in T_x\}$ induces a connected subtree of T , and for each edge vw of G there is a node $x \in V(T)$ such that $v, w \in T_x$. The *width* of a T -decomposition is the maximum, taken over the nodes $x \in V(T)$, of $|T_x| - 1$. The *treewidth* of a graph G is the minimum width of a tree decomposition of G . The *adhesion* of a tree decomposition $(T_x : x \in V(T))$ is $\max\{|T_x \cap T_y| : xy \in E(T)\}$. The *torso* of each node $x \in V(T)$ is the graph obtained from the induced subgraph $G[T_x]$ by adding a clique on $T_x \cap T_y$ for each edge $xy \in E(T)$ incident to x . Dujmovic et al [17] generalized Theorem 2 as follows.

Lemma 6 (Dujmovic et al [17]). *If a graph G has a tree decomposition with adhesion k such that every torso is nonrepetitively c -colorable, then G is nonrepetitively $c \cdot 4^k$ -colorable.*

For integers $c, d \geq 0$ a graph G has (c, d) -bounded degree if G contains at most c vertices with degree greater than d .

Lemma 7. *Every graph with (c, d) -bounded degree is nonrepetitively $c + (1 + o(1))d^2$ -colorable.*

Proof. Assign a distinct color to each vertex of degree at least d , and color the remaining graph by Theorem 1. For each vertex v of degree at least d , no other vertex is assigned the same color as v . Thus v is in no repetitively colored path. The result then follows from Theorem 1. $\square$

Dvořák [18] proved the following structure theorem for graphs excluding a strong immersion.

Theorem 8 (Dvořák [18]). *For every fixed graph H , there exists a constant k , such that every graph G that does not contain H as a strong immersion has a tree decomposition such that each torso is (k, k) -bounded degree.*

Lemmas 7 and 6 and Theorem 8 imply Theorem 4.

3 | WEAK IMMERSIONS

The proof of Theorem 4 gives no explicit bound on the constant k . In this section, we prove an explicit bound on the nonrepetitive chromatic number of graphs excluding a weak immersion. Theorem 3 follows from Lemma 6 and the following structure theorem of independent interest.

Theorem 9. *For every graph H with t vertices, every graph that does not contain H as a weak immersion has a tree decomposition with adhesion at most t^2 such that every torso has $(t, t^4 + 2t^2)$ -bounded degree.*

The starting point for the proof of Theorem 9 is the following structure theorem of Wollan [42]. For a tree T and graph G , a T -partition of G is a partition $(T_x \subseteq V(G) : x \in V(T))$ of $V(G)$ indexed by the nodes of T . Each set T_x is called a *bag*. Note that a bag may be empty. For each edge xy of a tree T , let $T(xy)$ and $T(yx)$ be the components of $T - xy$ where x is in $T(xy)$ and y is in $T(yx)$. For each edge $xy \in E(T)$, let $G(T, xy) := \bigcup\{T_z : z \in V(T(xy))\}$ and

$G(T, yx) := \bigcup \{T_z : z \in V(T(yx))\}$. Let $E(T, xy)$ be the set of edges in G between $G(T, xy)$ and $G(T, yx)$. The *adhesion* of a T -partition ($T_x : x \in V(T)$) is the maximum, taken over all edges xy of T , of $|E(T, xy)|$. For each node x of T , the *torso* of x (with respect to a T -partition) is the graph obtained from G by identifying $G(T, yx)$ into a single vertex for each edge xy incident to x (deleting resulting parallel edges and loops).

Theorem 10 (Wollan [42]). *For every graph H with t vertices, for every graph G that does not contain H as a weak immersion, there is a T -partition of G with adhesion at most t^2 such that each torso has (t, t^2) -bounded degree.*

Proof of Theorem 9. Let G be a graph that does not contain H as a weak immersion. Consider the T -partition ($T_x : x \in V(T)$) of G from Theorem 10.

Let T' be obtained from T by orienting each edge towards some root vertex. We now define a tree decomposition ($T_x^* : x \in V(T)$) of G . Initialize $T_x^* := T_x$ for each node $x \in V(T)$. For each edge vw of G , if $v \in T_x$ and $w \in T_y$ and z is the least common ancestor of x and y in T' , then add v to T_α^* for each node α on the $\overrightarrow{xz}$ path in T' , and add w to T_α^* for each node α on the $\overrightarrow{yz}$ path in T' . Thus each vertex $v \in T_x$ is in a sequence of bags that correspond to a directed path from x to some ancestor of x in T' . By construction, the endpoints of each edge are in a common bag. Thus ($T_x^* : x \in V(T)$) is a tree decomposition of G .

Consider a vertex $v \in T_x^* \cap T_y^*$ for some edge $\overrightarrow{xy}$ of T' . Then v is in $G(T, xy)$ and v has a neighbor w in $G(T, yx)$, implying $vw \in E(T, xy)$. Thus $|T_x^* \cap T_y^*| \leq |E(T, xy)| \leq t^2$. That is, the tree decomposition ($T_x^* : x \in V(T)$) has adhesion at most t^2 .

Let G_x^+ be the torso of each node $x \in V(T)$ with respect to the tree decomposition ($T_x^* : x \in V(T)$). That is, G_x^+ is obtained from $G[T_x^*]$ by adding a clique on $T_x^* \cap T_y^*$ for each edge xy of T . Our goal is to prove that G_x^+ has $(t, t^4 + 2t^2)$ -bounded degree.

Consider a vertex v of G_x^+ . Then v is in the bag corresponding to at most one child node z of x , as otherwise v would belong to a set of bags that do not correspond to a directed path in T' . Since ($T_x^* : x \in V(T)$) has adhesion at most t^2 , v has at most t^2 neighbors in $T_x^* \cap T_z^*$. For the same reason, if p is the parent of x , then v has at most t^2 neighbors in $T_x^* \cap T_p^*$. Thus the degree of v in G_x^+ is at most the degree of v in $G[T_x^*]$ plus $2t^2$. Call this property $(\star)$.

First consider the case that $v \notin T_x$. Let z be the node of T for which $v \in T_z$. Since $v \in T_x^*$, by construction, x is an ancestor of z . Let y be the node immediately before x on the $\overrightarrow{xz}$ path in T' . We now bound the number of neighbors of v in T_x^* . Say $w \in N_G(v) \cap T_x^*$. If w is in $G(T, xy)$ then let e_w be the edge vw . Otherwise, w is in $G(T, yx)$ and thus w has a neighbor u in $G(T, xy)$ since $w \in T_x^*$; let e_w be the edge wu . Observe that $\{e_w : w \in N_G(v) \cap T_x^*\} \subseteq E(T, xy)$, and thus $|\{e_w : w \in N_G(v) \cap T_x^*\}| \leq t^2$. Since $e_u \neq e_w$ for distinct $u, w \in N_G(v) \cap T_x^*$, we have $|N_G(v) \cap T_x^*| \leq t^2$. By $(\star)$, the degree of v in G_x^+ is at most $3t^2$.

Now consider the case that $v \in T_x$. Suppose further that v is not one of the at most t vertices of degree greater than t^2 in the torso Q of x with respect to the given T -partition. Suppose that in Q , v has d_1 neighbors in T_x and d_2 neighbors not in T_x (the identified vertices). So $d_1 + d_2 \leq t^2$. Consider a neighbor w of v in $G[T_x^*]$ with $w \notin T_x$. Then $w \in G(T, yx)$ for some child y of x . For at most d_2 children y of x , there is a neighbor of v in $G(T, yx)$. Furthermore, for each child y of x , v has at most t^2 neighbors in $G(T, yx)$ since the T -partition has adhesion at most t^2 . Thus v has degree at most $d_1 + d_2 t^2 \leq t^4$ in $G[T_x^*]$. By $(\star)$, v has degree at most $2t^2 + t^4$ in G_x^+ .

Since $3t^2 \leq t^4 + 2t^2$, the torso G_x^+ has $(t, t^4 + 2t^2)$ -bounded degree. $\square$

4 | EXCLUDING A TOPOLOGICAL MINOR

Theorem 5 is an immediate corollary of Lemma 6 and the following structure theorem of Dvořák [18] that extends Theorem 8.

Theorem 11 (Dvořák [18]). *Let H be a fixed planar graph that has a planar embedding with all the vertices of H with degree at least 4 on a single face. Then there exists a constant k , such that every graph G that does not contain H as a topological minor has a tree decomposition such that each torso has (k, k) -bounded degree.*

While Theorem 11 is not explicitly stated in Dvořák [18], we now explain that it is in fact a special case of Theorem 3 in Dvořák [18]. This result provides a structural description of graphs excluding a given topological minor in terms of the following definition. For a graph H and surface Σ , let $\text{mf}(H, \Sigma)$ be the minimum, over all possible embeddings of H in Σ , of the minimum number of faces such that every vertex of degree at least 4 is incident with one of these faces. By assumption, for our graph H and for every surface Σ , we have $\text{mf}(H, \Sigma) = 1$. In this case, Theorem 3 of Dvořák [18] says that for some integer $k = k(H)$, every graph G that does not contain H as a topological minor is a clique sum of (k, k) -bounded degree graphs. It immediately follows that G has the desired tree decomposition. See Corollary 1.4 in Liu and Thomas [34] for a closely related structure theorem.

The following natural open problem arises from This study: Do graphs excluding a fixed planar graph as a topological minor have bounded nonrepetitive chromatic number? And what is the structure of such graphs?

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ORCID

David R. Wood  <http://orcid.org/0000-0001-8866-3041>

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